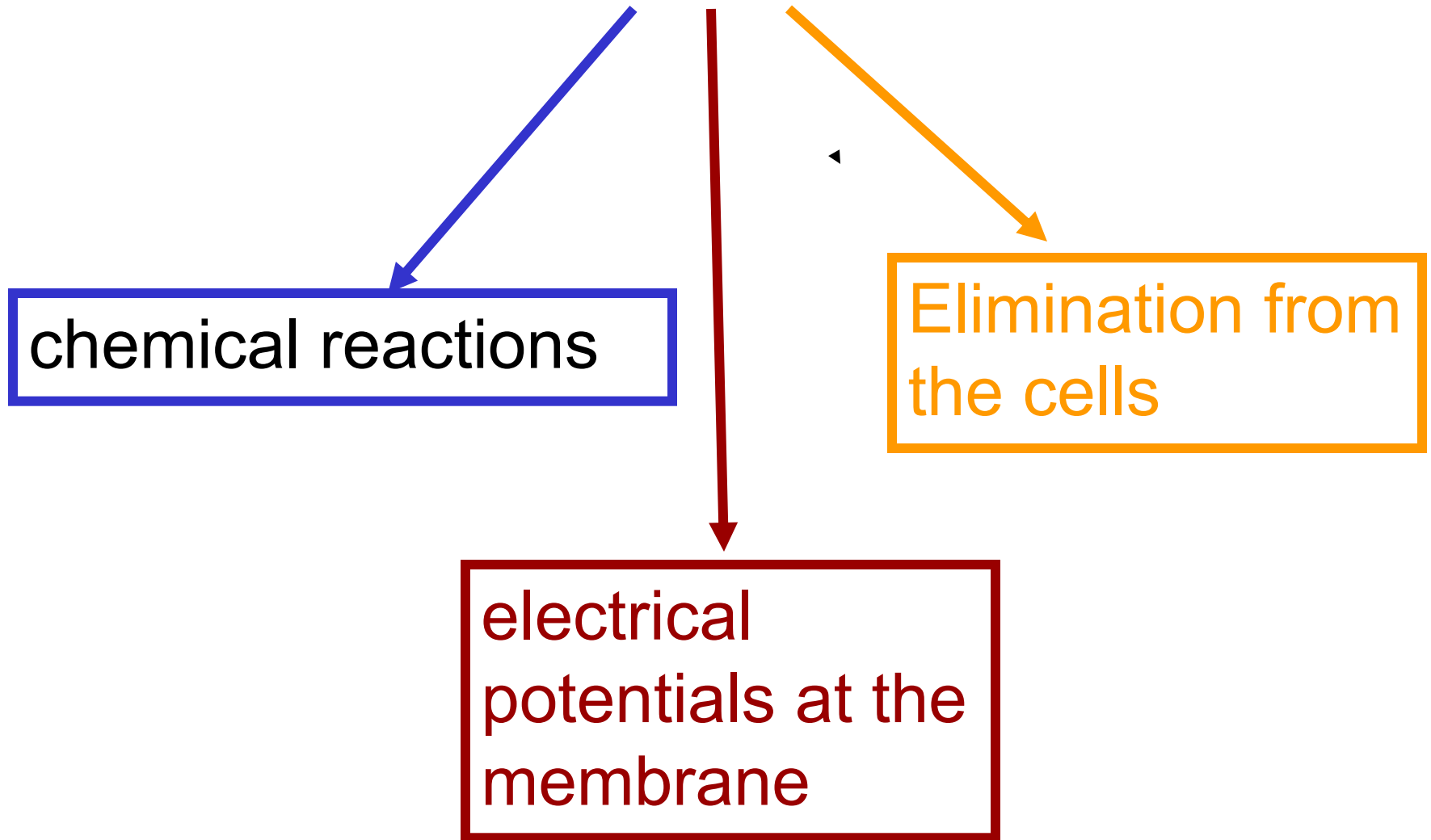


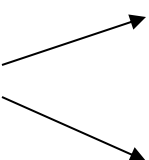
# **Cell membrane Transport across**

Dr katek balapala

# Substances transported for

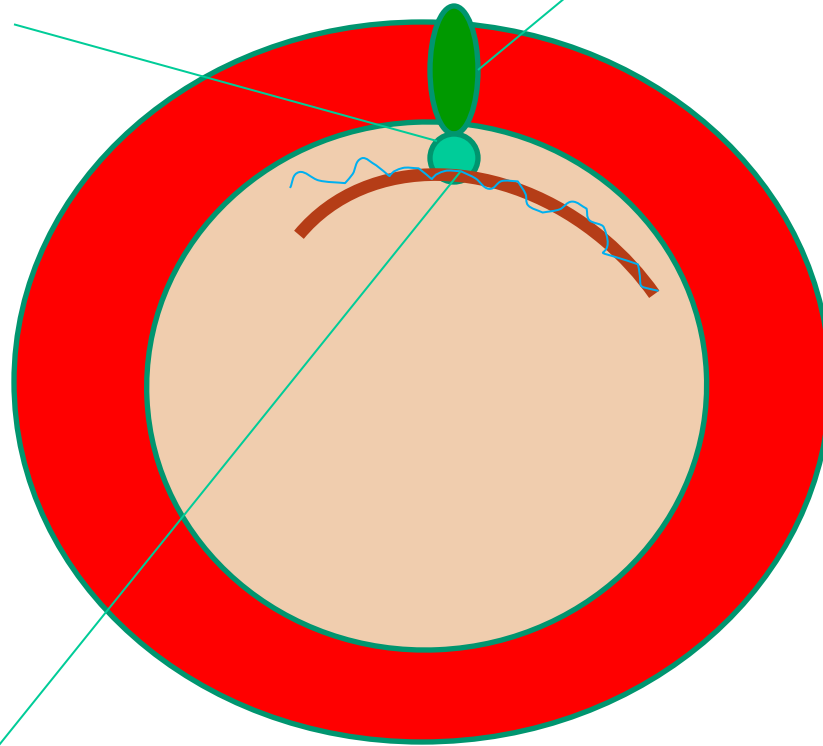


# Cell Membrane

- surrounds entire cell and cell organelles
- Fluid in nature – movement of molecules
- Phospholipid bilayer – head – polar/hydrophilic  
tail – nonpolar/hydrophobic
- Proteins 
  - Integral –carrier & channel
  - Peripheral-receptors & antigen

Peripheral protein  
(ankyrin)

Intrigal protein  
Band 3



cytoskelatal protein  
spectrin

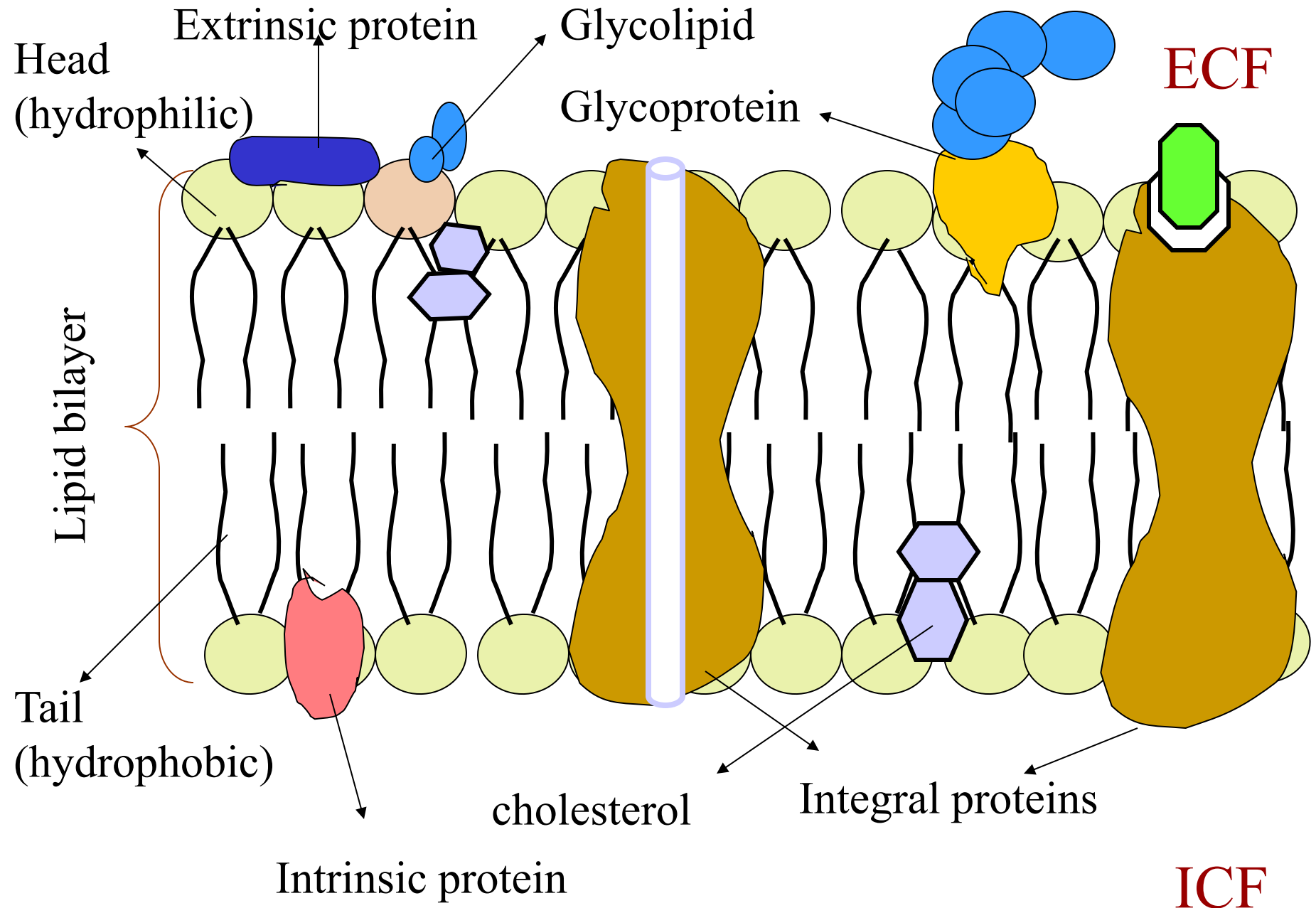
Fluid mosaic – membrane proteins like iceberg in sea –floating in sea of phospholipids which also moves

Translational diffusion –rate many micrometers/sec.  
Long chain & saturated fatty acids make membrane stiff. More kinks in unsaturated fatty acids-more fluidity

Cholesterol helps to maintain fluidity over wide range of temp.

Regional asymmetry of membrane

Transverse asymmetry



# Functions of cell membrane

- Acts as semi permeable barrier –(selective)
  - Maintains difference in composition of ICF & ECF & fluid in various organelles
  - Protects cell from toxic substances
  - Excretion of waste products
  - Transport of nutrients
- Receives signals from the outside
  - Chemical signals
  - Electrical signals
- Site for attachment to the neighboring cells

# Transport across cell membrane

## Transport Mechanisms



```
graph TD; A[Transport Mechanisms] --> B[Passive]; A --> C[Active]
```

### Passive

- Simple diffusion
- Facilitated diffusion
- Filtration
- Osmosis
- dialysis

### Active

- Primary active transport
- Secondary active transport
- Endo/Exocytosis



# Methods of transport

## Passive

## Active

Diffusion

Osmosis

Filtration

Dialysis

Simple

facilitated

- Lipid bilayer

- Protein channels

  - Leaky channels

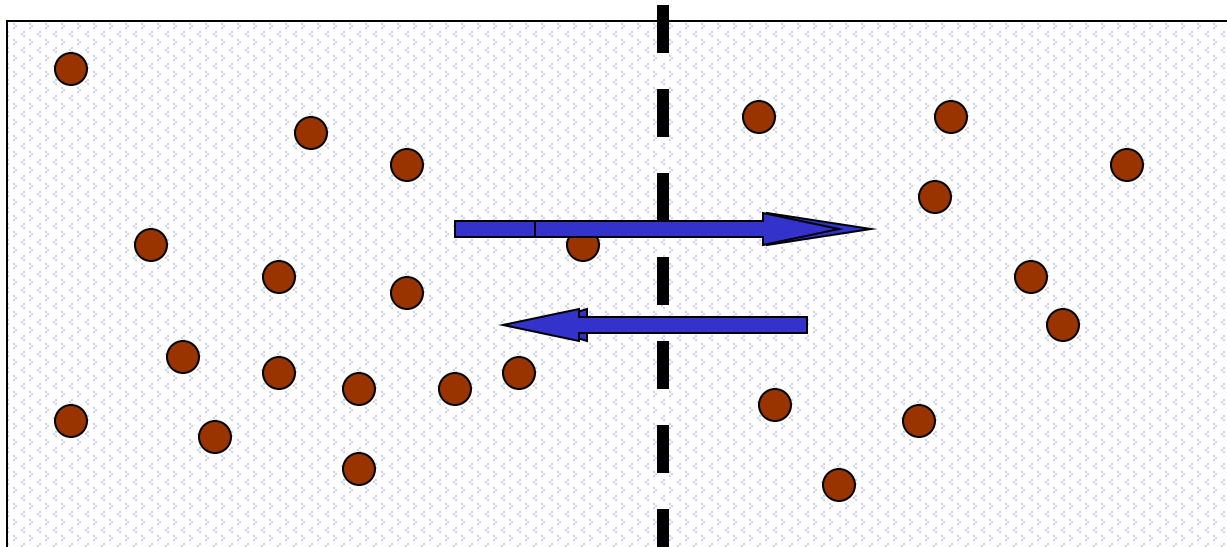
  - Gated channels

    - voltage gated

    - Ligand gated

## Simple diffusion -

Movement of molecules from higher concentration to lower concentration till equilibrium is reached



## .Diffusion can takes place through:

### a) Lipid bilayer

#### i) Lipid soluble substances-

O<sub>2</sub>,Co<sub>2</sub>,alcohol, steriods etc

#### ii) Lipid insoluble – water (through spaces bet lipid mol) urea, sugar

(less or no permeability)

#### iii) Electrolytes – impermeable

– charge on fatty acid chain

- Hydrated forms are larger

.

b) Protein Channels → Open/leaky – Na<sup>+</sup> channels, K<sup>+</sup> channels

Gated – channels open under specific conditions

Ligand gated

Voltage gated

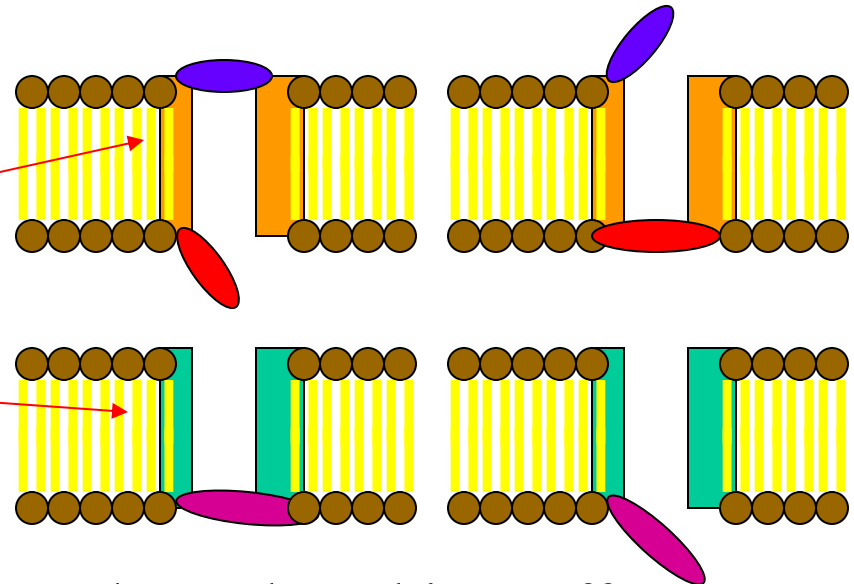
Na<sup>+</sup>

Na<sup>+</sup>,

K<sup>+</sup>

K<sup>+</sup>

Ca<sup>++</sup>,



Mutation of ionic channels produce channelopathies –affecting muscle and brain – paralysis or convulsions

# Factors affecting rate of diffusion

- Lipid solubility
  - Molecular size & wt.
  - Temperature
- Molecular**
- Thickness of membrane
  - Surface area
- Membrane related**
- Concentration gradient
  - Pressure gradient
  - Electrical gradient
- Gradients**

Imp.

## Fick's law of diffusion –

$$Q \propto \frac{\Delta C \cdot P \cdot A}{MW \cdot \Delta X}$$

$Q$  = net rate of diffusion

$\Delta C$  = conc. gradient of a substance

$P$  = permeability of membrane to the sub.

$A$  = surface area of a membrane

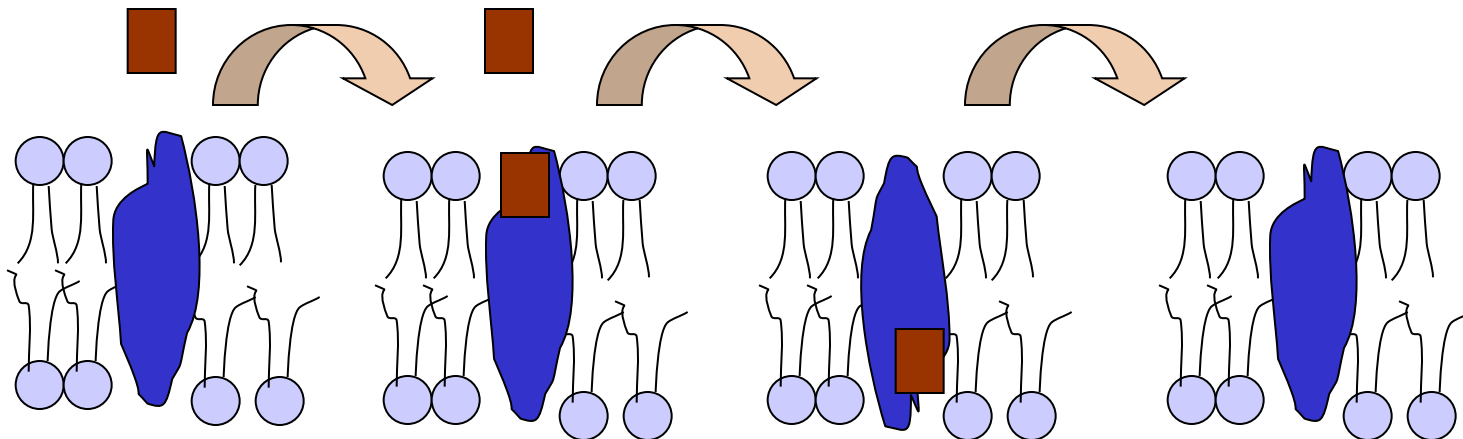
$MW$  = molecular wt. of sub.

$\Delta X$  = thickness or distance

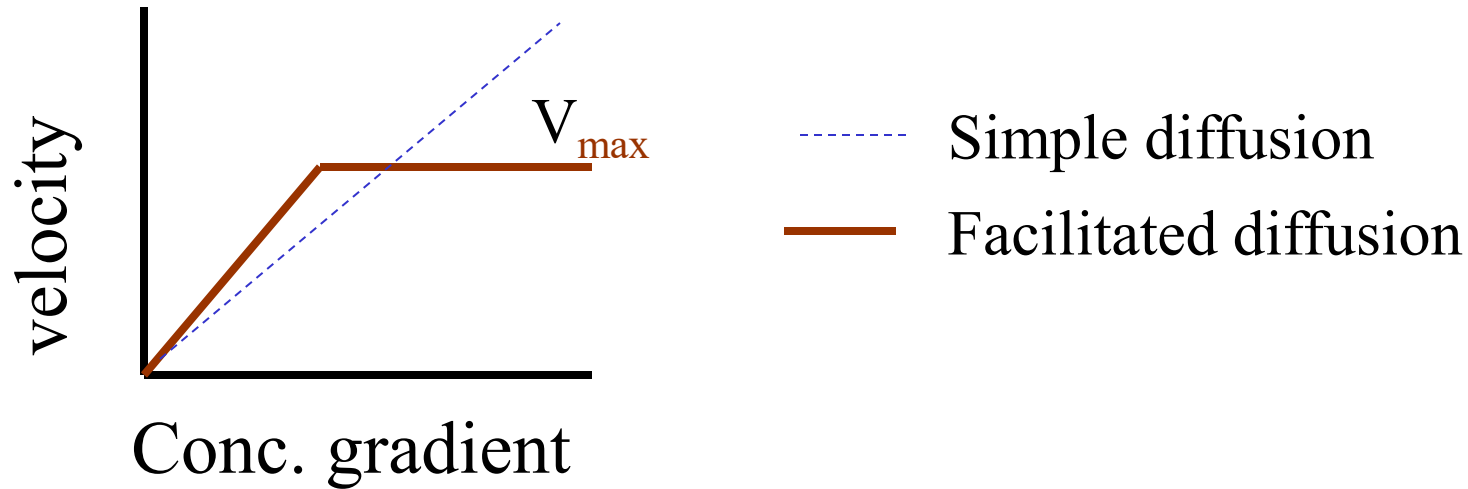
## II. Facilitated diffusion :

- for larger water soluble mols.
- type of passive transport
- along the conc. Gradient
- carrier mediated transport
- receptor site on one side

### Mechanism



- Rate of transport –  $V_{\max}$

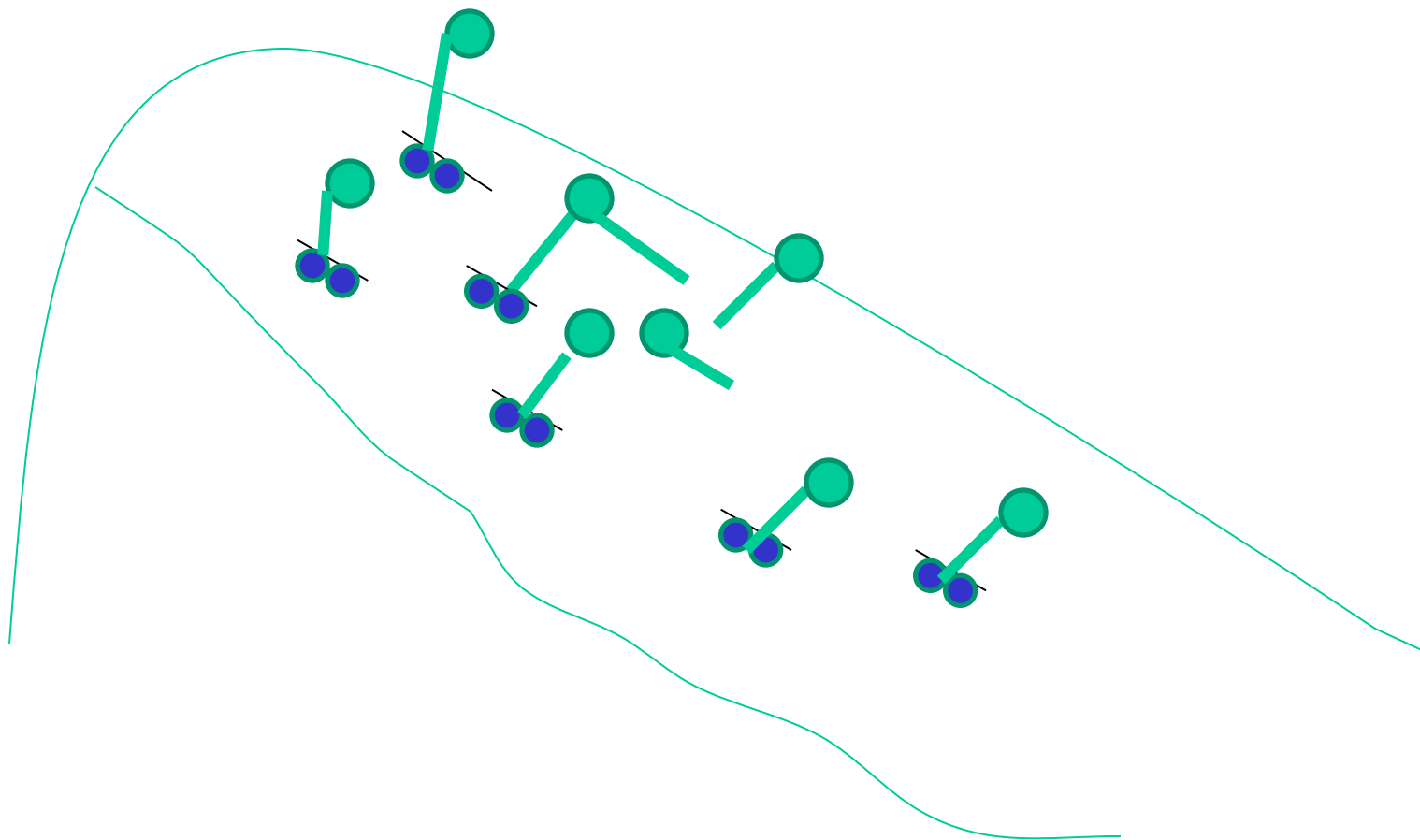


Initially, rate is directly proportional to conc. gradient

Till it reaches  $V_{\max}$  ( limitation because of no. of carrier mols. & rate of conformational change)

Hormonal regulation by changing #of carriers.



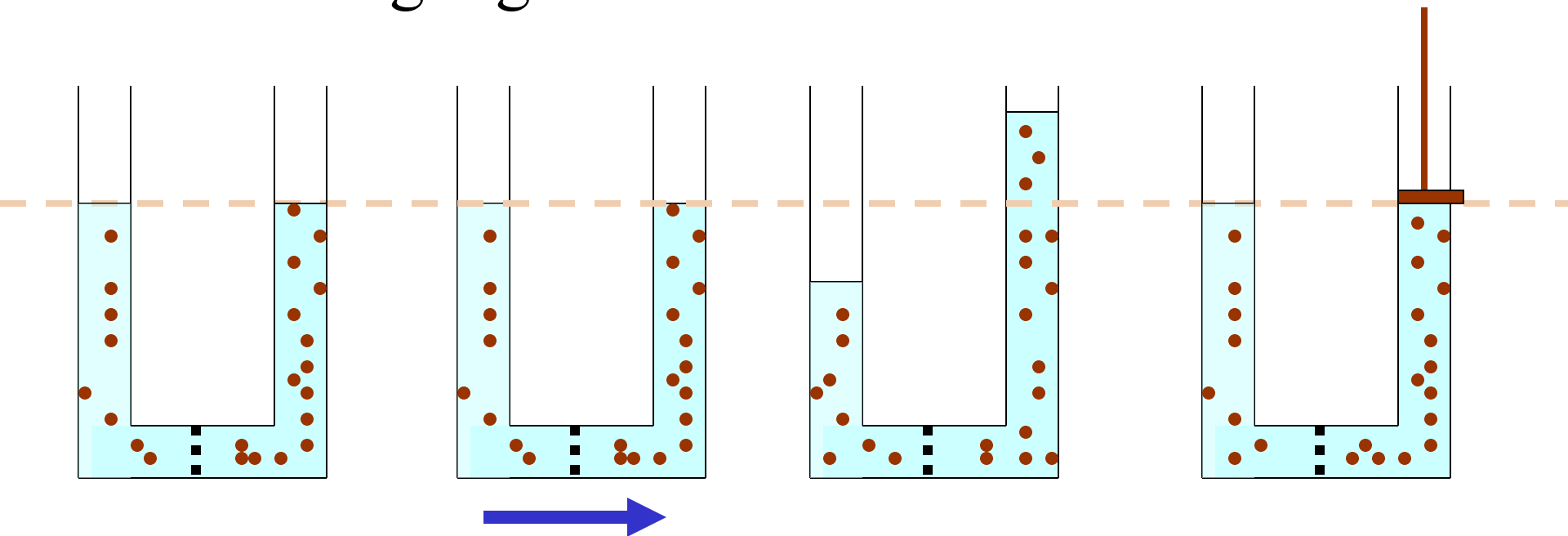


## - Peculiarities of carrier mediated transport –

- specificity,
- competitive or noncompetitive inhibition –  
phloridzin for glucose
- saturation,
- blocking of receptor
- $V_{\max}$
- Examples – transport of glucose, amino acids, galactose, etc. in the peripheral cells or counter transport of  $\text{Cl}^-$  and  $\text{HCO}_3^-$  in renal tubules

### III. Osmosis & osmotic pressure—

when two solutions of different concentrations are separated by a semi permeable membrane (impermeable to solute and permeable to water) water mols. diffuse from solution having less conc. To the sol. having higher conc.



**Osmotic pressure** is the minimum pressure applied on the solution with high conc. which prevents osmosis.

- depends upon total no. of particles of dissolved solutes rather than type of the particles

**Osmols or mOsmols** – expresses conc. of  
osmotically active particles

1 osmol = total no. of particles in gram molecular  
wt. of non diffusible substance per kg. of water

## Applied -

### Isotonic, hypotonic & hypertonic solutions

**Isotonic solution** – fluids having osmolarity same as that of plasma ( 290 mOsmols ) . Red cells suspended in such solution do not shrink or swell. ( 0.9 % NaCl, 5% glucose )

In **Hypotonic soln.** RBCs swell and hemolysis may occur.

In **hypertonic solution** RBCs shrink because water moves out.

# Diffusion potential or Equilibrium potential - E

Potential generated across the cell membrane in the presence of **non diffusible** ions in one compartment.

Magnitude of potential developed can be calculated by Nernst equation.

## Nernst equation -

Equilibrium potential or diffusion potential (E)

$$= \pm 61 \log \frac{\text{Conc. inside}}{\text{Conc. outside}}$$

$$E_K = -94 \text{ mV}$$

$$E_{Na} = +61 \text{ mV}$$

$$E_{Cl} = -90 \text{ mV}$$

Goldmann-Hodgkin's equation =

$$-61 \log \frac{C_{Na_i} \cdot P_{Na} + C_{K_i} \cdot P_{Na} + C_{Cl_o} \cdot P_{Cl_o}}{C_{Na_o} \cdot P_{Na} + C_{K_o} \cdot P_{Na} + C_{Cl_o} \cdot P_{Cl_i}}$$

## The Goldman-Hodgkin-Katz equation

$$V_m = \frac{RT}{F} \ln \left( \frac{p_K[K^+]_o + p_{Na}[Na^+]_o + p_{Cl}[Cl^-]_i}{p_K[K^+]_i + p_{Na}[Na^+]_i + p_{Cl}[Cl^-]_o} \right)$$

- **$V_m$**  is the membrane potential. This equation is used to determine the resting membrane potential in real cells, in which  $K^+$ ,  $Na^+$ , and  $Cl^-$  are the major contributors to the membrane potential. Note that the unit of  $V_m$  is the Volt. However, the membrane potential is typically reported in millivolts (mV). If the channels for a given ion ( $Na^+$ ,  $K^+$ , or  $Cl^-$ ) are closed, then the corresponding relative permeability values can be set to zero. For example, if all  $Na^+$  channels are closed,  $p_{Na} = 0$ .
- **$R$**  is the universal gas constant (8.314 J.K<sup>-1</sup>.mol<sup>-1</sup>).
- **$T$**  is the temperature in Kelvin (K = °C + 273.15).
- **$F$**  is the Faraday's constant (96485

[https://www.physiologyweb.com/calculators/ghk\\_equation\\_calculator.html](https://www.physiologyweb.com/calculators/ghk_equation_calculator.html)



## IV. Filtration

Filtration is a process in which fluid along with solutes passes through a membrane due to difference in pressures on both sides.

e.g. Filtration at capillary

Capillary hydrostatic pressure – 28mm Hg

Interstitial fluid hydrostatic pressure - -2mm Hg

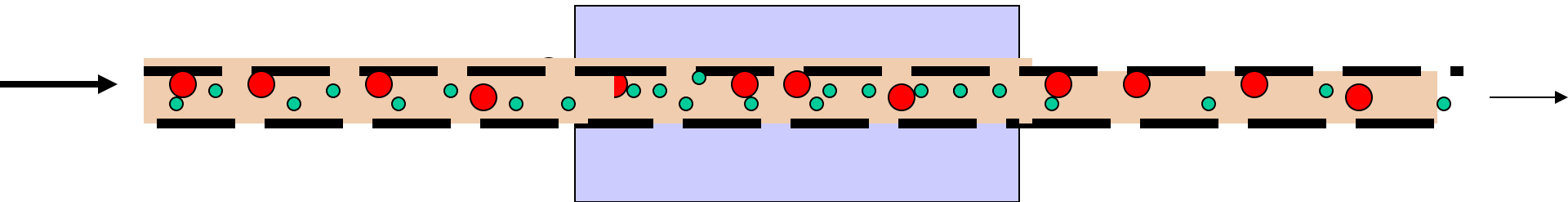
Colloidal osmotic pressure - 25mm Hg

Net Filtration pressure =  $28 - (-2 + 25) = 5 \text{ mm Hg}$

## V. Dialysis –

separation of larger dissolved particles from smaller particles

It is used for elimination of waste products in the blood in case of renal failure.



# Active transport

- Primary active transport
- Secondary active transport
- Endocytosis
  - Pinocytosis
  - Phagocytosis
- Exocytosis

# Peculiarities of active transport

- 1) Carrier mediated transport
- 2) Rapid rate of transport
- 3) Transport takes place against electrochemical gradient ( uphill )
- 4) Expenditure of energy by transport protein which incorporates ATPase activity

- 5) Carrier protein shows specificity, saturation  
competitive inhibition, blocking
- 6) Substances transported –  $\text{Na}^+$  ,  $\text{K}^+$  ,  $\text{H}^+$  ,  $\text{Cl}^-$  ,  $\text{I}^-$  ,  
Glucose, Amino acids

# I. Primary active transport –

Examples -  $\text{Na}^+$  -  $\text{K}^+$  pump,  $\text{Ca}^{++}$  pump

$\text{H}^+$ - $\text{K}^+$  pump

- Inner surface of carrier mol. has ATPase which is activated by attachment of specific ions and causes hydrolysis of ATP molecule
- Energy released from ATP causes conformational change in the carrier which transports ions to the opposite side.

## a) $\text{Na}^+ - \text{K}^+$ pump- electrogenic pump

- Attachment of  $2\text{K}^+$  on outer side &  $3\text{Na}^+$  on inner side

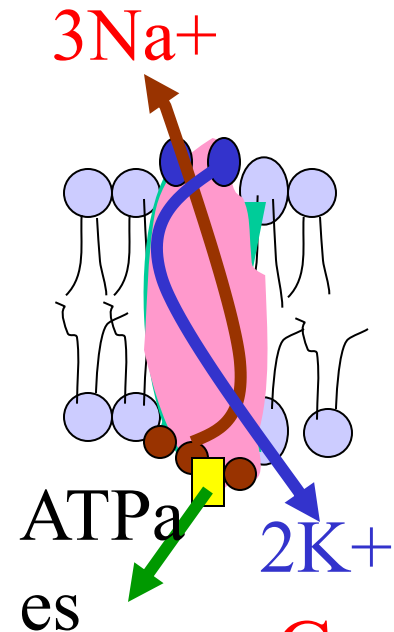
↓  
Activation of ATPase

↓  
Conformational change

↓  
Efflux of  $3\text{Na}^+$  & influx of  $2\text{K}^+$

↓  
Creates high  $\text{K}^+$  conc. & - vity inside the cell

Helps in maintaining cell volume



**Na-K pump** is one of the major energy using process in the body & accounts for a large part of basal metabolism.

## Regulators of Na-K pump –

- Increased amount of cellular Na conc.
- Thyroid hormones increase pump activity by more # of Na-K ATPase mol
- Aldosterone also increases # of pumps
- DOPamine inhibits pump
- Insulin increases pump activity
- Oubain or Digitalis inhibits ATPase (used when weakness of cardiac muscle –maintains Ca conc. In ICF of cardiac muscle



- **Ca<sup>++</sup> pump** –

present in the membrane of ER,  
mitochondria and cell membrane

- involves uniport carrier

- helps to maintain low Ca<sup>++</sup>conc. in ICF

## II. Secondary active transport

Active transport depending upon conc.

gradient of  $\text{Na}^+$  from ECF to ICF created by utilization of energy

\_ carrier does not have ATPase activity

Substance is transported along with  $\text{Na}^+$

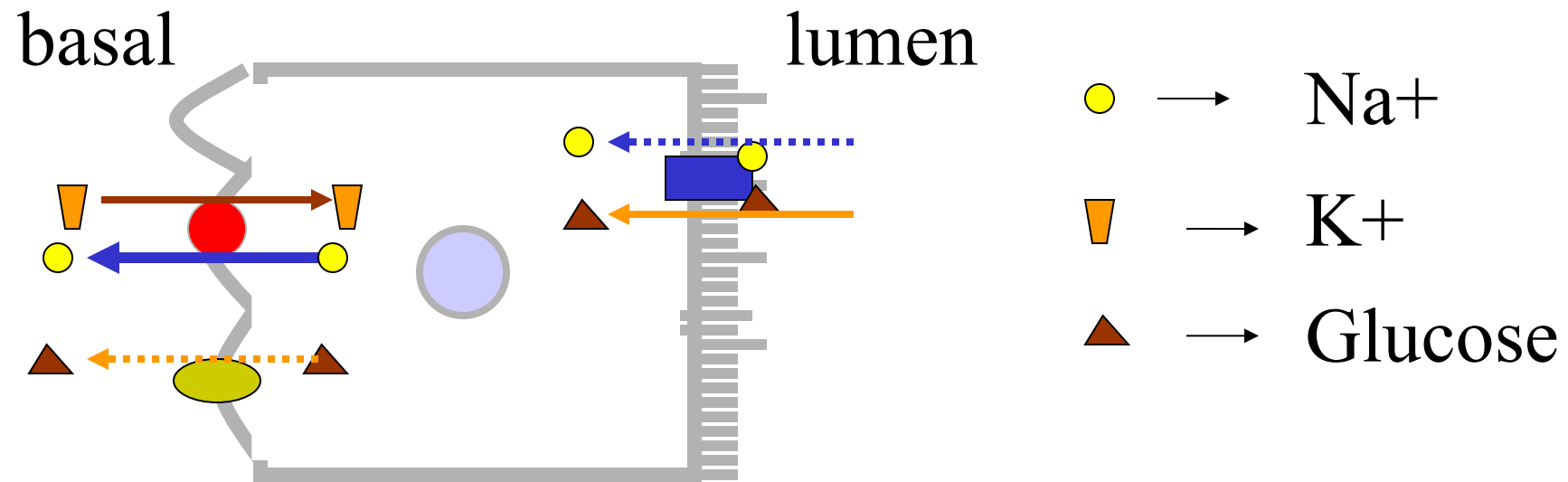
( $\text{Na}$  increases affinity of carrier for gl.)

$\text{Na}^+$  is transported only when glucose mol. is attached

Examples – a) Reabsorption of glucose & amino acids in PCT & Intestinal mucosa – **Co-transport mechanism**

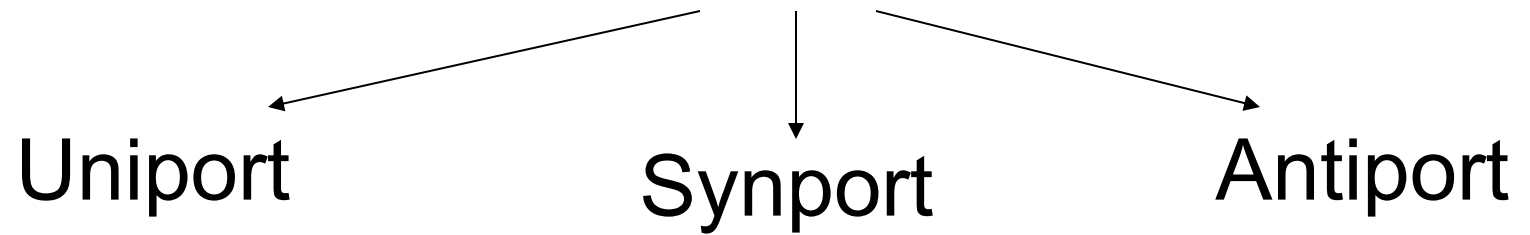
b)  $H^+$  secretion by tubular epithelium  
– **counter transport mechanism**

c) In heart Na-K ATPase indirectly affects Ca transport. – antiport in the membrane of cardiac muscle exchanges intracellular Ca for extracellular Na



- Na<sup>+</sup> - K<sup>+</sup> pump on basal side
- Electrochemical gradient for Na<sup>+</sup> on luminal side
- Carrier mediated transport (SGLT-1) of Na<sup>+</sup> along with glucose ( or amino acid ) through the apical membrane
  - Transport of glucose by facilitated diffusion ( GLUT-2 ) through basal side

# Types of transporters



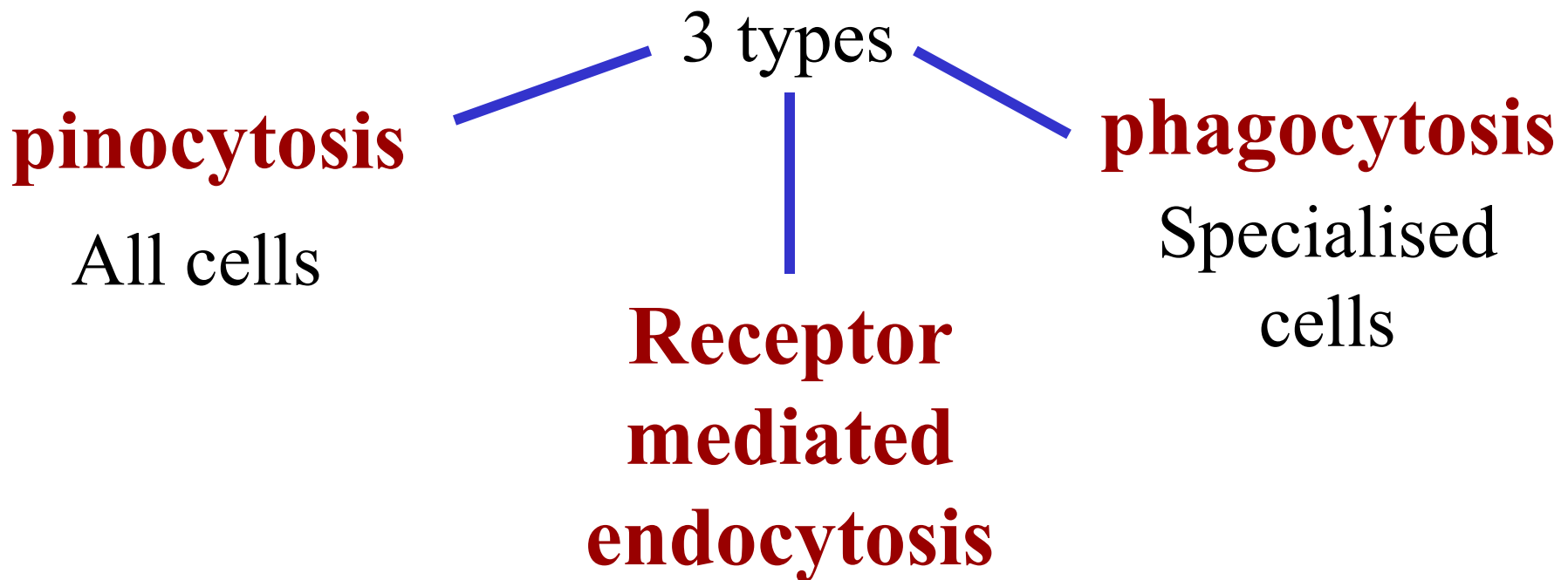
**uniporter** is the glucose transporter (GLUT) in found in erythrocytes (referred to as GLUT1 to separate from other mammalian glucose transporters). This allows glucose to enter the cell via facilitated diffusion

Symport = the  $\text{H}^+$ /oligopeptide transporter (PepT), found in the small intestine, couples the downhill movement of  $\text{H}^+$  across the plasma membrane to the uphill **transport** of dipeptides and tripeptides into the cell against a concentration gradient.

Antiport =

Na/Ca exchanger

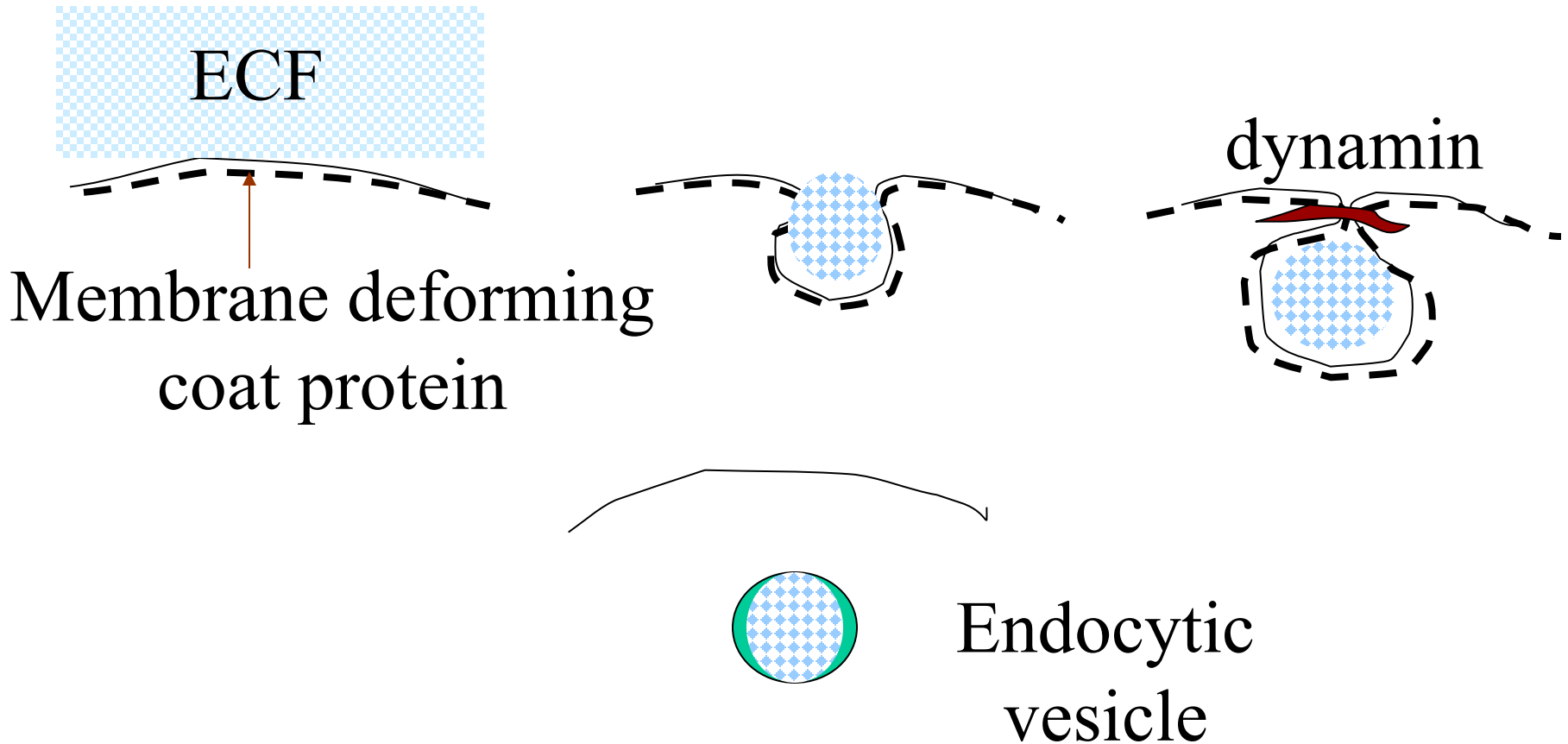
Extracellular material to be tackled by lysosomes is brought into the cell by  
**endocytosis**



Requires ATPase, Ca, microfilaments



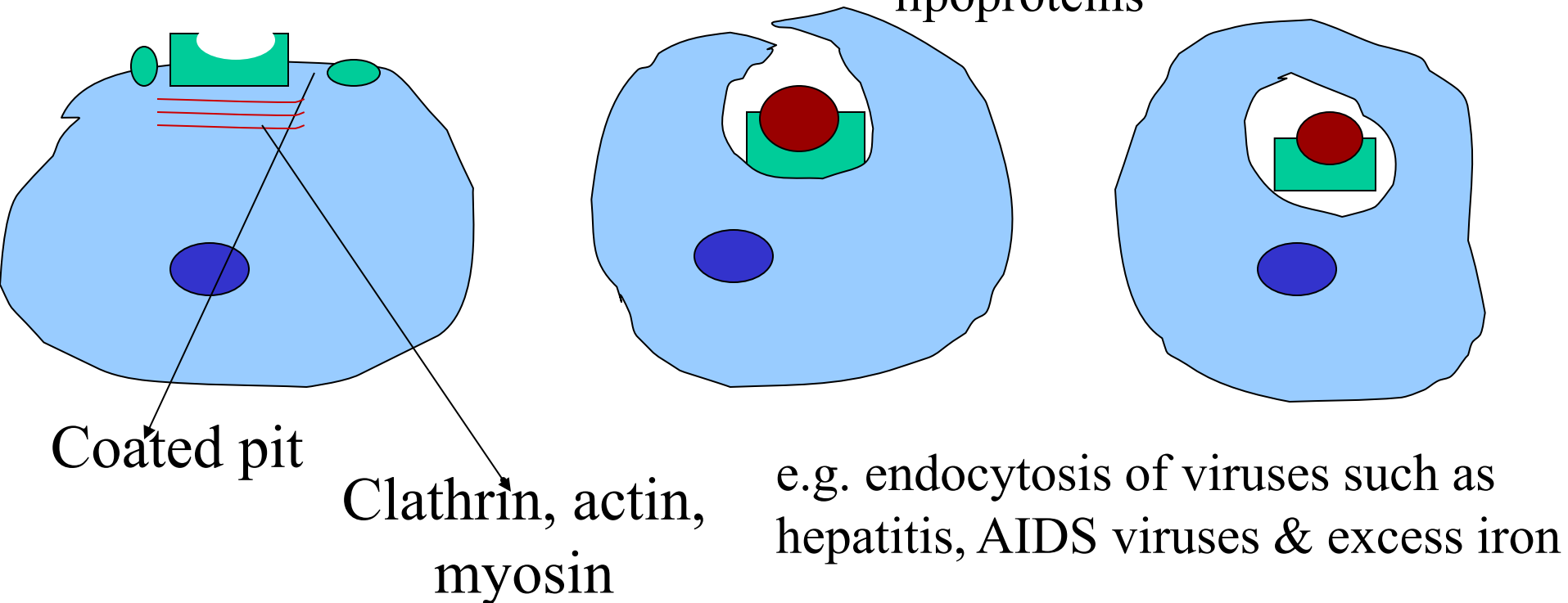
# Pinocytosis



**B. Receptor mediated endocytosis** – highly selective process to import imp. specific large molecules. Requires energy &  $\text{Ca}^{++}$ .



e.g. endocytosis of low density lipoproteins



## C. Phagocytosis

- Internalization of large multimolecular particles, bacteria, dead tissues by specialized cells e.g. certain types of w.b.c.s ( Professional phagocytes)
- The material makes contact with the cell membrane which then invaginates.

bacterium

Pseudopodia

internalization

Phagosome

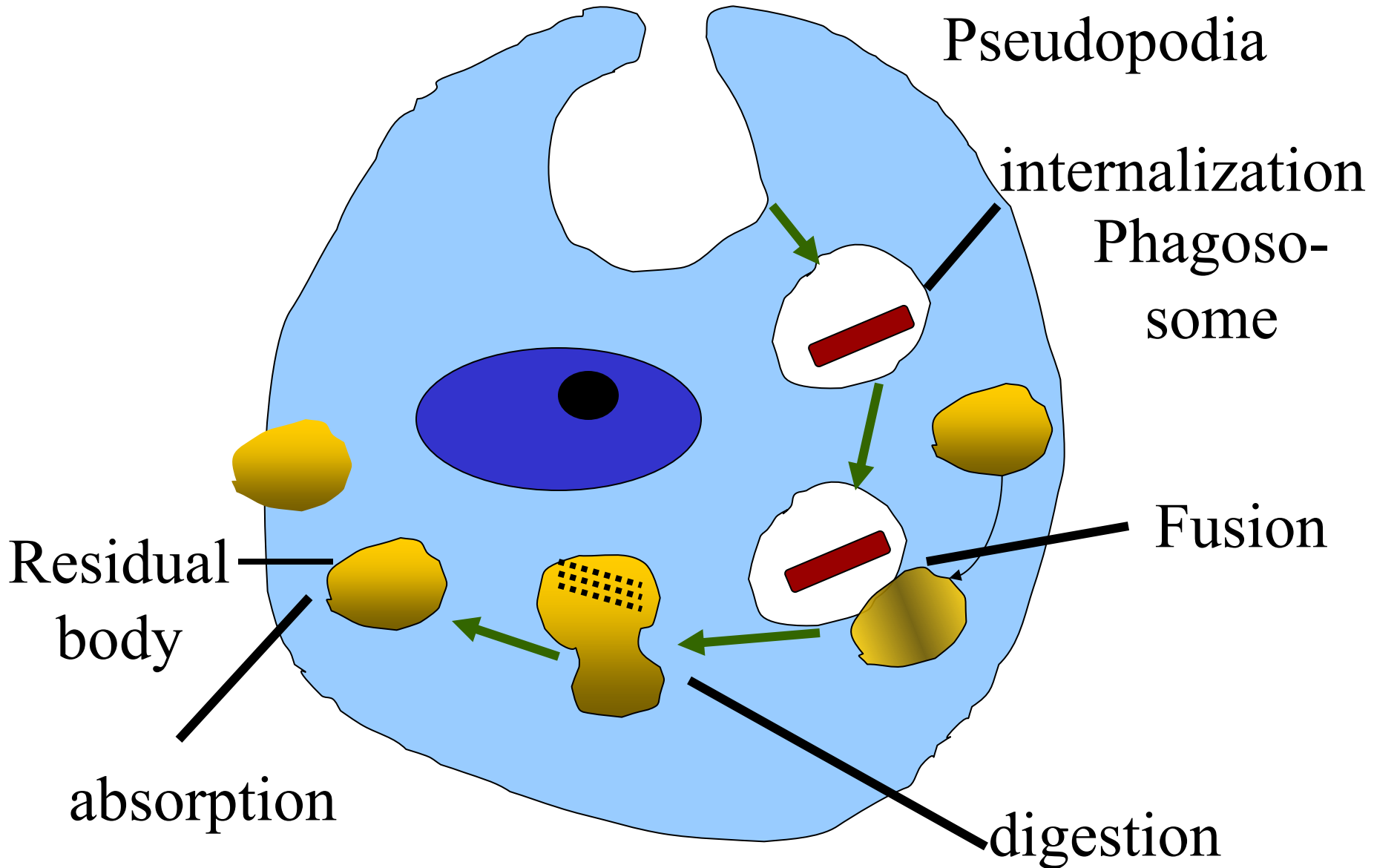
Fusion

Residual  
body

absorption

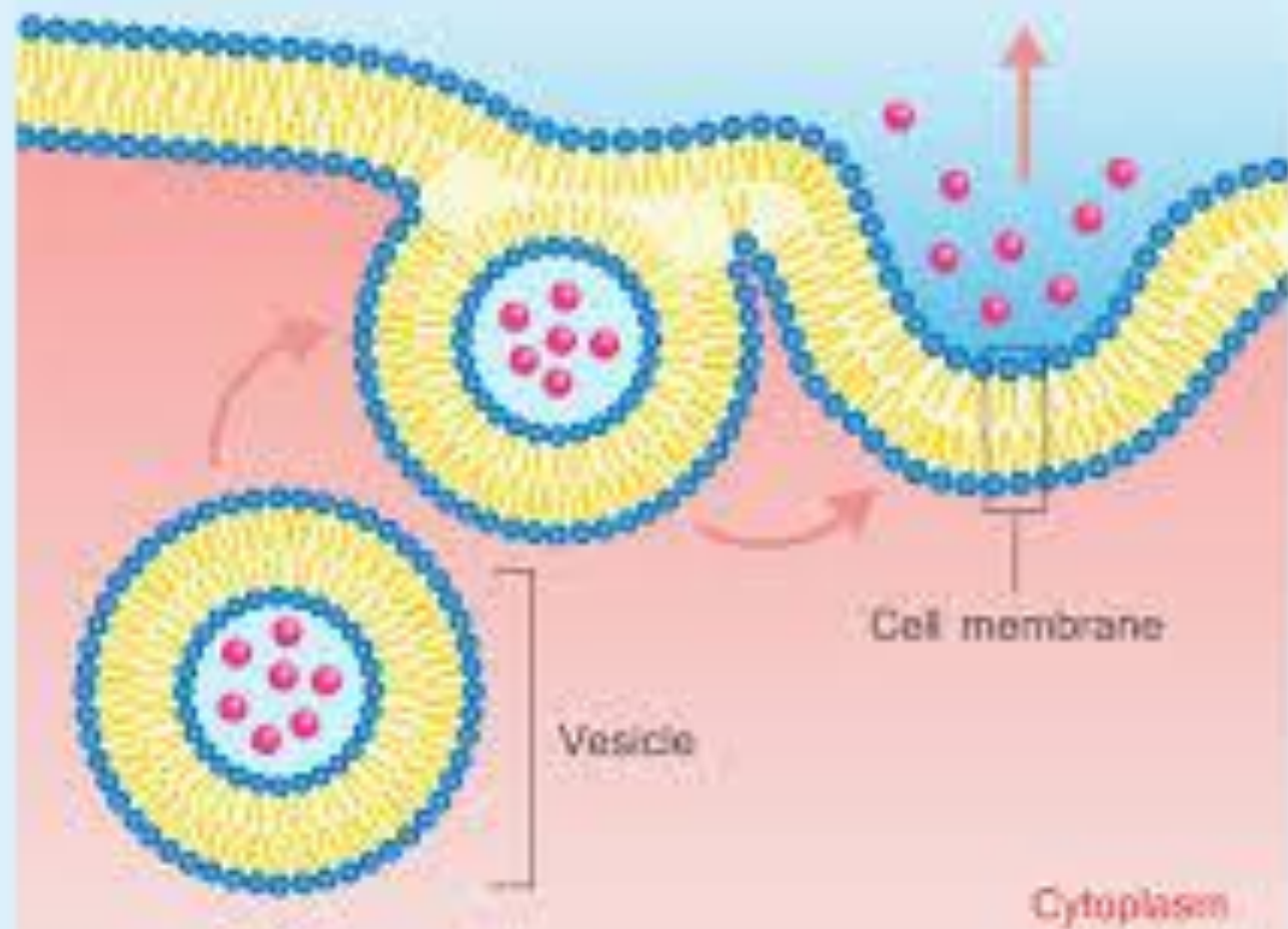
digestion

**Phagocytosis**



Exocytosis is a form of active transport and bulk transport in which a cell transports molecules out of the cell by secreting them through an energy-dependent process

Outside the cell



## Passive transport

- No expenditure of energy molecules
- Takes place along conc., electrical, & pressure gradient
- Carrier may or may not be required
- Rate is proportional to conc. difference

## Active transport

- Expenditure of energy mol. ( ATP )
- Can take place against conc. Gradient
- Carrier is always required
- Rate is proportional to availability of carrier & energy. ( $V_{\max}$ )

## Simple Diffusion

- Passive transport
- For small molecules
- No carrier required
- Rate of transport is directly proportional to conc. gradient
- Examples –
  - Lipid soluble –  
O<sub>2</sub>, CO<sub>2</sub>, alcohol
  - Lipid insoluble –  
urea, Na<sup>+</sup>, K<sup>+</sup>

## Facilitated Diffusion

- Passive transport
- For large molecules
- Carrier mediated
- Initially rate is proportional to conc. gradient till  $V_{\max}$   
(saturation of carriers)
- Examples –  
glucose, amino acids



